Email Service Architecture

The email service handles all outbound and inbound email communication for the ticketing system.

Components

Outbound Email (SMTP)

Triggering Events

Event	Email Type	Template Key
Ticket created	Confirmation	`ticketCreated`
Ticket status changed	Update	`ticketUpdated`
Ticket resolved	Resolution	`ticketResolved`
Ticket assigned	Assignment	`ticketAssigned`

Event	Email Type	Template Key
New comment added	Notification	`commentAdded`
Overdue SLA alert	Alert	`slaBreached`

Email Templates

Email templates are stored in the `EmailTemplate` model and managed through the admin panel.

Default templates include:

- **Ticket Created:** Confirms receipt and provides ticket number.
- **Ticket Updated:** Notifies requester of status or priority changes.
- **Ticket Resolved:** Announces resolution and requests feedback.

Configuration

SMTP settings are stored in `SystemSettings` and can be updated via `/api/settings`:

```
EMAIL_SERVICE=gmail
EMAIL_USER=itsupport@kuccps.ac.ke
EMAIL_PASSWORD=app-password
EMAIL_FROM="KUCCPS IT Support <itsupport@kuccps.ac.ke>"
SUPPORT_EMAIL=itsupport@kuccps.ac.ke
```

Inbound Email Monitoring (IMAP)

How It Works

1. The backend polls the IMAP mailbox at the configured interval.
2. New emails are parsed to extract subject, body, and attachments.
3. Emails are matched to existing tickets using the subject line or references.
4. Ticket content is updated, or new comments are added.
5. Notifications are sent to assigned staff.

Configuration

```
IMAP_HOST=imap.gmail.com
IMAP_PORT=993
IMAP_USER=itsupport@kuccps.ac.ke
IMAP_PASSWORD=app-password
IMAP_TLS=true
IMAP_POLL_INTERVAL=5
```

Enabling/Disabling

Email monitoring starts automatically on server boot if both `EMAIL_USER` and `IMAP_USER` are configured. It can be toggled via the admin panel.

Email Configuration Model

The `EmailConfig` model tracks configured mailboxes:

Field	Purpose
`provider`	Email provider name (e.g., `gmail`)
`email`	Unique mailbox address
`isActive`	Whether this config is active
`lastChecked`	Last time the mailbox was polled

Security Considerations

1. **TLS Enforcement:** Use `IMAP_TLS_REJECT_UNAUTHORIZED=true` in production.
2. **App Passwords:** For Gmail, use app passwords instead of account passwords.
3. **Firewall Rules:** Ensure the server can reach SMTP (port 587/465) and IMAP (port 993) hosts.
4. **Rate Limits:** The backend applies rate limiting to public endpoints to prevent abuse.

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